

A Proof of the Fundamental Theorem of Finitely Generated Abelian Groups

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Notation. All matrices and vectors will be assumed to have integer elements.

Definition. Let $v_1, \dots, v_n \in \mathbb{Z}^d$. Define

$$\mathbb{Z}\text{pan}(v_1, \dots, v_n) = \{\alpha_1 v_1 + \dots + \alpha_n v_n : \alpha_1, \dots, \alpha_n \in \mathbb{Z}\}.$$

Proposition 1. For $v_1, \dots, v_n \in \mathbb{Z}^d$, $\mathbb{Z}\text{pan}(v_1, \dots, v_n) \leq \mathbb{Z}^d$.

Proof. Suppose that $\alpha = \alpha_1 v_1 + \dots + \alpha_n v_n, \beta = \beta_1 v_1 + \dots + \beta_n v_n \in \mathbb{Z}\text{pan}(v_1, \dots, v_n)$. Then,

$$\alpha + \beta = (\alpha_1 + \beta_1)v_1 + \dots + (\alpha_n + \beta_n)v_n \in \mathbb{Z}\text{pan}(v_1, \dots, v_n)$$

and

$$-\alpha = (-\alpha_1)v_1 + \dots + (-\alpha_n)v_n \in \mathbb{Z}\text{pan}(v_1, \dots, v_n).$$

■

Definition. Let

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix} \in \mathcal{M}_{m \times n}(\mathbb{Z}).$$

Define the group associated to A by

$$\mathcal{G}(A) = \mathbb{Z}^m / \mathbb{Z}\text{pan}(v_1, \dots, v_n).$$

Definition. Let A be a matrix. We define $\mathbb{Z}$ row operations as the following:

1. Adding one row to another
2. Swapping two rows
3. Multiplying a row by -1

We define $\mathbb{Z}$ column operations as:

1. Adding one column to another
2. Swapping two columns
3. Multiplying a column by -1

Remark 2. If $k \in \mathbb{Z}$, we can add k times a row of A to another row. If $k \in \mathbb{Z}^+$, this is because we can use the first $\mathbb{Z}$ row move k times. If $k \in \mathbb{Z}^-$, this is because we can use the third $\mathbb{Z}$ row move followed by the first $\mathbb{Z}$ row move k times followed by the third $\mathbb{Z}$ row move again. The same is true of columns.

Proposition 3. Let A be a matrix. Then $\mathcal{G}(A)$ is preserved under $\mathbb{Z}$ column operations.

Proof. Let the columns of A be $v_1, \dots, v_n$. It suffices to show that $\mathbb{Z}\text{pan}(v_1, \dots, v_n)$ is preserved. We show this for each $\mathbb{Z}$ column operation.

1. Suppose WLOG that v_2 is added to v_1 . Then

$$\alpha_1 v_1 + \dots + \alpha_n v_n = \alpha_1(v_1 + v_2) + (\alpha_2 - \alpha_1)v_2 + \dots + \alpha_n v_n,$$

so that $\mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(v_1 + v_2, v_2, \dots, v_n)$. Also,

$$\alpha_1(v_1 + v_2) + \alpha_2 v_2 + \dots + \alpha_n v_n = \alpha_1 v_1 + (\alpha_1 + \alpha_2)v_2 + \dots + \alpha_n v_n,$$

so that $\mathbb{Z}\text{pan}(v_1 + v_2, v_2, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n)$.

2. Suppose WLOG that v_1 and v_2 are swapped. Then,

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = \alpha_2 v_2 + \alpha_1 v_1 + \dots + \alpha_n v_n,$$

so $\mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(v_2, v_1, \dots, v_n)$. Also,

$$\alpha_2 v_2 + \alpha_1 v_1 + \dots + \alpha_n v_n = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n,$$

so that $\mathbb{Z}\text{pan}(v_2, v_1, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n)$.

3. Suppose WLOG that v_1 is multiplied by -1 . Then,

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = (-\alpha_1)(-v_1) + \alpha_2 v_2 + \dots + \alpha_n v_n,$$

so that $\mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(-v_1, v_2, \dots, v_n)$. Also,

$$\alpha_1(-v_1) + \alpha_2 v_2 + \dots + \alpha_n v_n = (-\alpha_1)v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n,$$

so that $\mathbb{Z}\text{pan}(-v_1, v_2, \dots, v_n) \subseteq \mathbb{Z}\text{pan}(v_1, v_2, \dots, v_n)$. ■

Proposition 4. *Let A be a matrix. Then $\mathcal{G}(A)$ is preserved under $\mathbb{Z}$ row operations.*

Proof. Suppose that we are working with

$$\mathbb{Z}^m / \mathbb{Z}\text{pan} \left(\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} v_{n,1} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix} \right)$$

We find an isomorphism between the $\mathbb{Z}^m / \mathbb{Z}\text{pan}$'s obtained by performing $\mathbb{Z}$ row operations.

1. Suppose WLOG that the first two rows are added together. Define

$$\phi : \underbrace{\mathbb{Z}^m / \mathbb{Z}\text{pan} \left(\overbrace{\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}}^{v_1}, \dots, \overbrace{\begin{pmatrix} v_{n,1} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix}}^{v_n} \right)}_{\mathbb{Z}\text{pan}_1} \rightarrow \mathbb{Z}^m / \underbrace{\mathbb{Z}\text{pan} \left(\begin{pmatrix} v_{1,1} + v_{1,2} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} v_{n,1} + v_{n,2} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix} \right)}_{\mathbb{Z}\text{pan}_2}$$

by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1 \mapsto \begin{pmatrix} a_1 + a_2 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2$. We claim that ϕ is a well-defined isomorphism. First, if

$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \equiv \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = b \pmod{\mathbb{Z}\text{pan}_1}$, we can write $a - b = \sum_{i=1}^n \alpha_i v_i$ for $\alpha_i \in \mathbb{Z}$. Then,

$$\phi(a) = \begin{pmatrix} a_1 + a_2 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} = \begin{pmatrix} b_1 + \sum_{i=1}^n \alpha_i v_{i,1} + b_2 + \sum_{i=1}^n \alpha_i v_{i,2} \\ b_2 + \sum_{i=1}^n \alpha_i v_{i,2} \\ \vdots \\ b_m + \sum_{i=1}^n \alpha_i v_{i,m} \end{pmatrix} \equiv \begin{pmatrix} b_1 + b_2 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \pmod{\mathbb{Z}\text{pan}_2} = \phi(b),$$

so ϕ is well-defined. We can find $\psi : \mathbb{Z}^m / \mathbb{Z}\text{pan}_2 \rightarrow \mathbb{Z}^m / \mathbb{Z}\text{pan}_1$ by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2 \mapsto \begin{pmatrix} a_1 - a_2 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1$, which is well-defined by a similar argument. Clearly ϕ and ψ are inverses, so ϕ is bijective. Finally,

$$\begin{aligned} \phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right) &= \phi \left(\begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{pmatrix} \right) = \begin{pmatrix} a_1 + b_1 + a_2 + b_2 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{pmatrix} = \begin{pmatrix} a_1 + a_2 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} b_1 + b_2 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \\ &= \phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \right) + \phi \left(\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right), \end{aligned}$$

so ϕ is a homomorphism. Hence, ϕ is a well-defined isomorphism.

2. Suppose WLOG that we swap the first two rows. Define

$$\phi : \mathbb{Z}^m / \underbrace{\mathbb{Z}\text{pan} \left(\begin{pmatrix} \overbrace{v_1} \\ v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} \overbrace{v_n} \\ v_{n,1} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix} \right)}_{\mathbb{Z}\text{pan}_1} \rightarrow \mathbb{Z}^m / \underbrace{\mathbb{Z}\text{pan} \left(\begin{pmatrix} v_{1,2} \\ v_{1,1} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} v_{n,2} \\ v_{n,1} \\ \vdots \\ v_{n,m} \end{pmatrix} \right)}_{\mathbb{Z}\text{pan}_2}$$

by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1 \mapsto \begin{pmatrix} a_2 \\ a_1 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2$. We claim that ϕ is a well-defined isomorphism. First, if

$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \equiv \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = b \pmod{\mathbb{Z}\text{pan}_1}$, we can write $a - b = \sum_{i=1}^n \alpha_i v_i$ for $\alpha_i \in \mathbb{Z}$. Then,

$$\phi(a) = \begin{pmatrix} a_2 \\ a_1 \\ \vdots \\ a_m \end{pmatrix} = \begin{pmatrix} b_2 + \sum_{i=1}^n \alpha_i v_{i,2} \\ b_1 + \sum_{i=1}^n \alpha_i v_{i,1} \\ \vdots \\ b_m + \sum_{i=1}^n \alpha_i v_{i,m} \end{pmatrix} \equiv \begin{pmatrix} b_2 \\ b_1 \\ \vdots \\ b_m \end{pmatrix} = \phi(b),$$

so ϕ is well-defined. We can find $\psi : \mathbb{Z}^m / \mathbb{Z}\text{pan}_2 \rightarrow \mathbb{Z}^m / \mathbb{Z}\text{pan}_1$ by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2 \mapsto \begin{pmatrix} a_2 \\ a_1 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1$,

which is well-defined by the same argument. Clearly ϕ and ψ are inverses, so ϕ is bijective. Finally,

$$\phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right) = \phi \left(\begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{pmatrix} \right) = \begin{pmatrix} a_2 + b_2 \\ a_1 + b_1 \\ \vdots \\ a_m + b_m \end{pmatrix} = \begin{pmatrix} a_2 \\ a_1 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} b_2 \\ b_1 \\ \vdots \\ b_m \end{pmatrix} = \phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \right) + \phi \left(\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right),$$

so ϕ is a homomorphism. Hence, ϕ is a well-defined isomorphism.

3. Suppose WLOG that the first row is multiplied by -1 . Define

$$\phi : \mathbb{Z}^m / \underbrace{\mathbb{Z}\text{pan} \left(\underbrace{\begin{pmatrix} v_1 \\ v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} v_n \\ v_{n,1} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix}}_{\mathbb{Z}\text{pan}_1} \right)}_{\mathbb{Z}\text{pan}_1} \rightarrow \mathbb{Z}^m / \underbrace{\mathbb{Z}\text{pan} \left(\begin{pmatrix} -v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,m} \end{pmatrix}, \dots, \begin{pmatrix} -v_{n,1} \\ v_{n,2} \\ \vdots \\ v_{n,m} \end{pmatrix} \right)}_{\mathbb{Z}\text{pan}_2}$$

by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1 \mapsto \begin{pmatrix} -a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2$. We claim that ϕ is a well-defined isomorphism. First, if

$a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \equiv \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = b \pmod{\mathbb{Z}\text{pan}_1}$, we can write $a - b = \sum_{i=1}^n \alpha_i v_i$ for $\alpha_i \in \mathbb{Z}$. Then,

$$\phi(a) = \begin{pmatrix} -a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} = \begin{pmatrix} -(b_1 + \sum_{i=1}^n \alpha_i v_{i,1}) \\ b_2 + \sum_{i=1}^n \alpha_i v_{i,2} \\ \vdots \\ b_m + \sum_{i=1}^n \alpha_i v_{i,m} \end{pmatrix} \equiv \begin{pmatrix} -b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = \phi(b),$$

so ϕ is well-defined. We can find $\psi : \mathbb{Z}^m / \mathbb{Z}\text{pan}_2 \rightarrow \mathbb{Z}^m / \mathbb{Z}\text{pan}_1$ by $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_2 \mapsto \begin{pmatrix} -a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \mathbb{Z}\text{pan}_1$,

which is well-defined by the same argument. Clearly ϕ and ψ are inverses, so ϕ is bijective. Finally,

$$\begin{aligned} \phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right) &= \phi \left(\begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{pmatrix} \right) = \begin{pmatrix} -(a_1 + b_1) \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{pmatrix} = \begin{pmatrix} -a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} + \begin{pmatrix} -b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \\ &= \phi \left(\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} \right) + \phi \left(\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \right), \end{aligned}$$

so ϕ is a homomorphism. Hence, ϕ is a well-defined isomorphism. ■

Definition. Call an $m \times n$ matrix A *pseudo-diagonal* if it satisfies $A_{i,j} = 0$ whenever $i \neq j$. Call A *diagonal* if it is pseudo-diagonal and square. Given a pseudo-diagonal matrix A , call the *padded diagonal matrix* of A the matrix obtained by adding or deleting columns of 0's to make A square. Denote the padded diagonal matrix of A by $\mathcal{P}(A)$.

Proposition 5. *Let A be a matrix. Then*

$$\mathcal{G}(A) = \mathcal{G} \left(\begin{pmatrix} A & 0 \end{pmatrix} \right).$$

Proof. Adding or removing zero vectors does not affect the $\mathbb{Z}\text{pan}$. ■

Corollary 6. *If A is a pseudo-diagonal matrix, $\mathcal{G}(A) = \mathcal{G}(\mathcal{P}(A))$.*

Proposition 7. *Let*

$$A = \begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & \ddots & \\ & & & a_n \end{pmatrix}.$$

Then,

$$\mathcal{G}(A) \cong \prod_{i=1}^n \mathbb{Z}/a_i\mathbb{Z}.$$

Proof. Define $\phi : \mathbb{Z}^n \rightarrow \prod_{i=1}^n \mathbb{Z}/a_i\mathbb{Z}$ by

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mapsto (x_1 + a_1\mathbb{Z}, x_2 + a_2\mathbb{Z}, \dots, x_n + a_n\mathbb{Z}).$$

Clearly, ϕ is a surjective homomorphism. The kernel of ϕ is the set of vectors of the form

$$\begin{pmatrix} k_1 a_1 \\ k_2 a_2 \\ \vdots \\ k_n a_n \end{pmatrix} = k_1 \begin{pmatrix} a_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 0 \\ a_2 \\ \vdots \\ 0 \end{pmatrix} + \dots + k_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ a_n \end{pmatrix}$$

for $k_1, \dots, k_n \in \mathbb{Z}$. Thus,

$$\prod_{i=1}^n \mathbb{Z}/a_i\mathbb{Z} \cong \mathbb{Z}^n / \ker \phi = \mathbb{Z}^n / \mathbb{Z}\text{pan} \left(\begin{pmatrix} a_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ a_2 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ a_n \end{pmatrix} \right) = \mathcal{G}(A). ■$$

Definition. Let v and w be vectors. We define the *Euclidean Algorithm on the i th term on (v, w)* as the following algorithm:

1. If v_i or w_i is negative, multiply the corresponding vector by -1 .
2. If $w_i > v_i$, swap v and w .
3. If $w_i = 0$, stop.
4. Replace v with $v - w$.
5. Go to (2).

Proposition 8. Let $v, w \in \mathbb{Z}^k$. Then the *Euclidean Algorithm on the i th term on v and w* halts. Further, if the output after it halts is (v', w') , then $v'_i = \gcd(v_i, w_i)$ and $w'_i = 0$.

Proof. After the first step of the algorithm, we are guaranteed that v_i and w_i are nonnegative. We now prove that the algorithm halts with $v'_i = \gcd(v_i, w_i)$ and $w'_i = 0$ by induction on $v_i + w_i$. The base case is $v_i + w_i = 0$, in which case $v_i = w_i = 0$, so we halt with $v'_i = v_i = 0 = \gcd(0, 0)$ and $w'_i = w_i = 0$. Now suppose that, whenever $v_i + w_i \leq k$, we halt with $v'_i = \gcd(v_i, w_i)$ and $w'_i = 0$. Let $v_i + w_i = k + 1$, and suppose that $v_i \geq w_i$, since we would switch them otherwise. If $w_i = 0$, we terminate with $v'_i = v_i = \gcd(v_i, 0)$ and $w'_i = w_i = 0$. Otherwise, we subtract w from v to get new vectors $\begin{pmatrix} | \\ v_i - w_i \\ | \end{pmatrix}$ and $\begin{pmatrix} | \\ w_i \\ | \end{pmatrix}$. Since w_i is positive, the sum $(v_i - w_i) + w_i = v_i$ is less than $v_i + w_i$. By the inductive hypothesis, this will terminate with $v'_i = \gcd(v_i - w_i, w_i) = \gcd(v_i, w_i)$ and $w'_i = 0$. ■

Lemma 9. Let M be an $m \times n$ matrix with $m, n \geq 2$ of the form

$$M = \begin{pmatrix} a & 0 & 0 & \cdots & 0 \\ b & | & | & \cdots & | \\ 0 & v_1 & v_2 & \cdots & v_{n-1} \\ 0 & | & | & \cdots & | \end{pmatrix}$$

for $a > 0$. Then M can be put into the form

$$\begin{pmatrix} a' & \\ & A \end{pmatrix}$$

using $\mathbb{Z}$ row and $\mathbb{Z}$ column moves.

Proof. We prove this by induction on a . The base case is $a = 1$, in which case we can subtract b times the first row from the second row. Suppose that we can put any such matrix in the appropriate form using $\mathbb{Z}$ row and $\mathbb{Z}$ column moves for any $a \leq k$, and that $a = k + 1$. Suppose also that $b \neq 0$, as we are otherwise done. If $a \mid b$, we can subtract the first row b/a times from the second row, so suppose that $a \nmid b$. In particular, $\gcd(a, b) < a$. Use the Euclidean Algorithm on the First Term on $(v_1, v_2), (v_1, v_3), \dots, (v_1, v_{n-1})$ to get a matrix of the form

$$\begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ 0 & - & X \end{pmatrix}.$$

Apply the Euclidean Algorithm on the First Term on the first two rows of M . We now have a matrix of the form

$$\begin{pmatrix} \gcd(a, b) & d & 0 \\ 0 & e & 0 \\ 0 & - & X \end{pmatrix}.$$

Now, apply the Euclidean Algorithm on the First Term on the first two columns of M to get a matrix of the form

$$\begin{pmatrix} a' & 0 & 0 \\ f & g & 0 \\ | & - & X' \end{pmatrix}.$$

Finally, apply the Euclidean Algorithm on the First Term on the rows $(M_2, M_3), (M_2, M_4), \dots, (M_2, M_m)$. This gives us a new matrix of the form

$$\begin{pmatrix} a' & 0 & 0 & \cdots & 0 \\ b' & | & | & \cdots & | \\ 0 & v'_1 & v'_2 & \cdots & v'_n \\ 0 & | & | & \cdots & | \end{pmatrix}.$$

We have that $a' \leq \gcd(a, b) < a$. By the inductive hypothesis, we can use $\mathbb{Z}$ row and $\mathbb{Z}$ column operations to put this in the desired form. ■

Theorem 10. *Let A be a $m \times n$ matrix. Then, we can make A pseudo-diagonal using $\mathbb{Z}$ row and $\mathbb{Z}$ column operations.*

Proof. Firstly, if $m = n = 1$, we are done. If $m = 1$ and $n \neq 1$, we can repeatedly use the Euclidean Algorithm on the First Term on the rows to get all elements of A to be 0 but the first (i.e. to make A pseudo-diagonal). If $n = 1$ and $m \neq 1$, we can do the same on columns.

We now prove that we can make A pseudo-diagonal by induction on $m + n$. We have verified the base case of $m + n = 2$, so suppose that we can make A pseudo-diagonal for $m + n \leq k$ and that $m + n = k + 1$. If either m or n is 1, we know how to do this, so suppose they are both at least 2. Apply the Euclidean Algorithm on the First Term on the columns $(A^1, A^2), (A^1, A^3), \dots, (A^1, A^n)$. Then, apply the Euclidean Algorithm on the First Term on the rows $(A_2, A_3), \dots, (A_2, A_m)$. This gives a matrix that has $A_{1,i} = 0$ for $i \neq 1$ and $A_{i,1} = 0$ for $i \neq 1, 2$. By our lemma, we can make this into some

$$\begin{pmatrix} a & \\ & A' \end{pmatrix},$$

where A' is $(m-1) \times (n-1)$. By the inductive hypothesis, we can make A' into a pseudo-diagonal matrix using $\mathbb{Z}$ row and $\mathbb{Z}$ column operations. Performing those operations here, we can make A into a pseudo-diagonal matrix. ■

Corollary 11. *Let A be a matrix. Then, there are some $a_1, \dots, a_n$ with*

$$\mathcal{G}(A) \cong \prod_{i=1}^n \mathbb{Z}/a_i\mathbb{Z}.$$

Corollary 12. *Any sub $\mathbb{Z}$ space of $\mathbb{Z}^n$ is $\mathbb{Z}$ spanned by finitely many vectors.*

Proof. Suppose that some sub $\mathbb{Z}$ space were not $\mathbb{Z}$ spanned by finitely many vectors. Then, in particular, there would exist $n+1$ vectors that were $\mathbb{Z}$ linearly independent, so it suffices to show that this cannot happen. If we had $n+1$ vectors in $\mathbb{Z}^n$, we could put them into an $n \times (n+1)$ matrix A . By the above theorem, we can use $\mathbb{Z}$ row and $\mathbb{Z}$ column operations to turn this into a pseudo-diagonal matrix. In particular, the rightmost column of this matrix will be 0, and so the columns of this matrix are $\mathbb{Z}$ linearly dependent. Noticing that $\mathbb{Z}$ linear dependence and independence are preserved under $\mathbb{Z}$ row and $\mathbb{Z}$ column operations, this means that our original $n+1$ vectors were $\mathbb{Z}$ linearly dependent. ■

Lemma 13. *We can use $\mathbb{Z}$ row and $\mathbb{Z}$ column operations to turn $\begin{pmatrix} n & 0 \\ 0 & m \end{pmatrix}$ into $\begin{pmatrix} \gcd(n, m) & 0 \\ 0 & \text{lcm}(n, m) \end{pmatrix}$.*

Proof. If $m = n = 0$, we are done. Otherwise, one of m and n is nonzero, so that $\gcd(m, n) \neq 0$ in particular. Writing $\gcd(n, m) = k_1n + k_2m$,

$$\begin{aligned} \begin{pmatrix} n & 0 \\ 0 & m \end{pmatrix} &\xrightarrow{C_2 + k_1 C_1} \begin{pmatrix} n & k_1 n \\ 0 & m \end{pmatrix} \xrightarrow{R_1 + k_2 R_2} \begin{pmatrix} n & k_1 n + k_2 m \\ 0 & m \end{pmatrix} \rightarrow \begin{pmatrix} n & \gcd(n, m) \\ 0 & m \end{pmatrix} \\ &\xrightarrow{R_2 - \frac{m}{\gcd(n, m)} R_1} \begin{pmatrix} n & \gcd(n, m) \\ -\frac{mn}{\gcd(n, m)} & 0 \end{pmatrix} \xrightarrow{C_1 - \frac{n}{\gcd(n, m)} C_2} \begin{pmatrix} 0 & \gcd(n, m) \\ -\frac{mn}{\gcd(n, m)} & 0 \end{pmatrix} \\ &\xrightarrow{-R_2} \begin{pmatrix} 0 & \gcd(n, m) \\ \frac{mn}{\gcd(n, m)} & 0 \end{pmatrix} \xrightarrow{C_1 \leftrightarrow C_2} \begin{pmatrix} \gcd(n, m) & 0 \\ 0 & \frac{mn}{\gcd(n, m)} \end{pmatrix} \rightarrow \begin{pmatrix} \gcd(n, m) & 0 \\ 0 & \text{lcm}(n, m) \end{pmatrix}. \end{aligned}$$

Proposition 14. *Any diagonal matrix*

$$\begin{pmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_m \end{pmatrix}$$

can be put into the form

$$\begin{pmatrix} n_1 & & & \\ & n_2 & & \\ & & \ddots & \\ & & & n_m \end{pmatrix},$$

where $\gcd(d_1, \dots, d_m) = n_1 |n_2| \cdots |n_m| = \text{lcm}(d_1, \dots, d_m)$ using $\mathbb{Z}$ row and $\mathbb{Z}$ column operations. Such a matrix is said to be in Smith Normal Form.

Proof. We prove this by induction on m . The base case is when $m = 1$. In this case, $n_1 = d_1$ can be obtained and satisfies the necessary conditions. Assume that any $m \times m$ diagonal matrix can be put into this form. Suppose we have a diagonal matrix

$$\begin{pmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_{m+1} \end{pmatrix}.$$

Use the inductive hypothesis on the first m rows and columns to get a matrix

$$\begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & \ddots & \\ & & & a_m \\ & & & & d_{m+1} \end{pmatrix}$$

where $a_1 | \cdots | a_m$. Apply the lemma to the last 2 rows and columns to get

$$\begin{pmatrix} a_1 & & & & \\ & a_2 & & & \\ & & \ddots & & \\ & & & \gcd(a_m, d_{m+1}) & \\ & & & & \text{lcm}(a_m, d_{m+1}) \end{pmatrix},$$

and use the inductive hypothesis once again on the first m rows and columns to get

$$\begin{pmatrix} n_1 & & & & \\ & n_2 & & & \\ & & \ddots & & \\ & & & n_m & \\ & & & & n_{m+1} \end{pmatrix}.$$

We claim that this is in Smith Normal Form. We know that $n_1 | n_2 | \cdots | n_m$, so we must show that $n_m | n_{m+1}$. By the inductive hypothesis,

$$n_m = \text{lcm}(a_1, a_2, \dots, a_{m-1}, \gcd(a_m, d_{m+1})) = \text{lcm}(a_{m-1}, \gcd(a_m, d_{m+1})).$$

Now,

$$a_{m-1} | a_m | \text{lcm}(a_m, d_{m+1}) \quad \gcd(a_m, d_{m+1}) | \text{lcm}(a_m, d_{m+1}),$$

so

$$n_m = \text{lcm}(a_{m-1}, \gcd(a_m, d_{m+1})) \mid \text{lcm}(a_m, d_{m+1}) = n_{m+1}$$

by the definition of lcm. Hence, we have that $n_1 \mid n_2 \mid \cdots \mid n_m \mid n_{m+1}$. It remains to show that $n_1 = \gcd(d_1, \dots, d_m)$ and $n_{m+1} = \text{lcm}(d_1, \dots, d_m)$. We have that

$$\begin{aligned} n_1 &= \gcd(a_1, a_2, \dots, a_{m-1}, \gcd(a_m, d_{m+1})) = \gcd(a_1, \dots, a_{m-1}, a_m, d_{m+1}) \\ &= \gcd(a_1, d_{m+1}) = \gcd(\gcd(d_1, \dots, d_m), d_{m+1}) = \gcd(d_1, \dots, d_{m+1}) \end{aligned}$$

and that

$$n_{m+1} = \text{lcm}(a_m, d_{m+1}) = \text{lcm}(\text{lcm}(d_1, \dots, d_m), d_{m+1}) = \text{lcm}(d_1, \dots, d_{m+1}).$$

■

Theorem 15. *Let G be a finitely generated abelian group. Then, there are integers $n_1 \mid \cdots \mid n_n$ with*

$$G \cong \prod_{i=1}^n \mathbb{Z}/n_i\mathbb{Z}.$$

Proof. Let $\{g_1, \dots, g_n\}$ be the set of generators for G . Add the relations “ $g_i g_j = g_j g_i$ ” for all i, j , and write all other relations in the form $g_1^{b_1} g_2^{b_2} \cdots g_n^{b_n} = e$. Take vectors $(v_i)_{i \in I}$ given by $v_1 e_1 + v_2 e_2 + \cdots + v_n e_n$. Consider the homomorphism from $\mathbb{Z}^n$ to G given by sending e_i to g_i . The kernel of this map is all tuples of exponents for the elements of the generating set that give the identity. These can be written using the relations in our presentation, and so the kernel of the map is the $\mathbb{Z}$ span of the v_i . Now, this $\mathbb{Z}$ span is finitely generated, say by $w_1, \dots, w_k$. Putting these into a matrix A , we see that $G \cong \mathcal{G}(A) \cong \mathcal{G}(\mathcal{P}(A))$. Thus, there is a diagonal matrix A' with $G \cong \mathcal{G}(A')$. Putting A' into Smith Normal Form gives the desired result. ■